

## BRIDGEPULSE · REQUIREMENTS

# BridgePulse requirements

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## DOCUMENT CONTROL

|             |                                          |         |                |
|-------------|------------------------------------------|---------|----------------|
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| Status      | <b>Draft</b>                             | Date    | 2026-09-25     |
| Author      | Amish Chadha                             | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept            |         |                |
| Repository  | github.com/BoujeeEnjinia1701/bridgepulse |         |                |
| Source      | docs/03-requirements.md @ 520f6f8        |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                                                           |
|---------|------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.4     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); R2 method (load gating), R5 verification, R7 restated with an 11-byte reduced summary; status from BRP-CAL-001 v0.2 |
| 0.3     | 2026-09-25 | Amish Chadha | R12 redefined per BRP-DDR-001 D1; status of every requirement from BRP-CAL-001; interface assumptions for FieldNode                                              |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept estimates                                                                               |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                                                                                                                 |

## BridgePulse requirements

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These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with bridge owners or inspectors, and will be revised after co-design sessions (see BRP-PRB-001). Status is now judged by calculation in BRP-CAL-001 against the parametric model and the priced BOM; "met on paper" is not a verification by test.

**Reference installation.** A steel or concrete footbridge or short-span road bridge of 5 to 30 m span, monitored at midspan, in a climate from  $-20$  to  $+50$  °C, with LoRaWAN coverage and no mains power. The worked example is the 7 m steel footbridge in BRP-PRC-001.

**Changes in v0.3.** R12 is redefined to cost the BridgePulse-specific parts, with the FieldNode core costed in its own repo and the complete cost stated beside it (BRP-DDR-001, D1; decided by Amish, 2026-09-25: go with recommendation). No other target changes.

**Changes in v0.4 (BRP-DDR-002).** R2 keeps its target; its method now gates out the time when people are on the span, backed by a load regression in TwinKit, and is to be checked on recorded data before any change to the target. R5 keeps its target; verification adds an ice-point check, and the probe is now TMP1826 class. R7 is restated so that an 11-byte reduced summary is used wherever the data rate in use limits the payload to 11 bytes. Statuses follow BRP-CAL-001 v0.2.

Table 1. Requirements and status at TRL 3 (BRP-CAL-001 v0.2, Table 5).

| ID  | Requirement                                         | Target                                                                                                                                                                                                                                                                                         | Verification                                                                           | Status at TRL 3                                                                                                                                                     |
|-----|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Measure bridge acceleration                         | 3 axes; noise density 25 $\mu\text{g}/\text{VHz}$ or less; band 0.5 to 60 Hz                                                                                                                                                                                                                   | Datasheet; later bench noise test                                                      | Met by design (22.5 $\mu\text{g}/\text{VHz}$ ; 62.5 Hz filter corner)                                                                                               |
| R2  | Track natural frequencies                           | Every mode with a clear peak below 60 Hz, at least the first vertical mode; bin width 0.02 Hz or less per hourly record; repeatability 0.2 % ( $1\sigma$ ) or better over a day of steady temperature                                                                                          | Spectral and load-gating simulation (BRP-CAL-001, B); then recorded data               | At risk: with the time when people are on the span gated out, 0.015 % or less in simulation (1 to 20 crossings per record); ungated, walker mass gives 1.5 to 3.1 % |
| R3  | Flag structural change                              | Flag a sustained shift of 1 % or more in a tracked frequency, after temperature compensation, within 14 days; one false flag per bridge per year or fewer                                                                                                                                      | Detection statistics (BRP-CAL-001, C); then public benchmark data and a field baseline | Not verifiable at TRL 3: needs a daily residual of 0.75 % or less after temperature and load compensation                                                           |
| R4  | Measure strain                                      | 2 channels; range $\pm 1,000 \mu\epsilon$ ; resolution 2 $\mu\epsilon$ or better; drift 5 $\mu\epsilon$ per month or less after temperature compensation                                                                                                                                       | ADC noise calculation; later outdoor drift test                                        | Not verifiable at TRL 3 for drift; resolution (0.09 $\mu\epsilon$ RMS) and range met on paper                                                                       |
| R5  | Measure temperature                                 | Steel surface and shaded air; $\pm 0.5 \text{ }^\circ\text{C}$ or better, checked against a reference                                                                                                                                                                                          | Datasheet; ice-point check at 0 $^\circ\text{C}$ and CalRig chamber check              | Met on paper: TMP1826 class, $\pm 0.3 \text{ }^\circ\text{C}$ from $-40$ to $+105 \text{ }^\circ\text{C}$ on the datasheet check                                    |
| R6  | Stay within the FieldNode energy budget             | Average draw 30 mW or less at the FieldNode sensor port                                                                                                                                                                                                                                        | Power calculation; later current logging                                               | Met on paper (16.9 mW)                                                                                                                                              |
| R7  | Send and keep the data                              | One summary per hour of 48 bytes or less, within LoRaWAN regional limits at the data rate in use, with an 11-byte reduced summary (first frequency, its amplitude, two strain means, steel temperature, status) wherever the limit is 11 bytes; raw records kept on site for 12 months or more | Payload and airtime calculation; storage calculation                                   | Met on paper: 36 bytes, or 11 bytes at US915 DR0 and AS923 DR2 (AS923 limit assumed); 22 months on 32 GB                                                            |
| R8  | Fit without harming the structure                   | No drilling, welding or cutting; strain gauges only on an area of paint removed and recoated with the owner's permission; fitted by two trained people in 4 h or less from below or beside the deck                                                                                            | Fitting sequence estimate; later trial fit                                             | Not verifiable at TRL 3 for time (2.8 h estimate); no drilling or welding met by design                                                                             |
| R9  | Keep clear of what passes under and over the bridge | Nothing more than 15 mm below the girder soffit; nothing on the deck side except the FieldNode on a handrail post, with tamper-resistant fasteners                                                                                                                                             | Model check                                                                            | Met on paper (10 mm; hub inside the girder outline)                                                                                                                 |
| R10 | Survive outdoors                                    | Hub IP67, FieldNode IP65; operate from $-20$ to $+50 \text{ }^\circ\text{C}$ ; 5 years with one gauge recoat and no cell replacement                                                                                                                                                           | Datasheets and design review; later environmental checks                               | At risk: FieldNode heat (FND-CAL-001) and gauge coating life                                                                                                        |
| R11 | Protect privacy                                     |                                                                                                                                                                                                                                                                                                | Design review                                                                          | Met by design                                                                                                                                                       |

| ID  | Requirement                 | Target                                                                                                                 | Verification                             | Status at TRL 3                                     |
|-----|-----------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------|
|     |                             | No cameras or microphones; only structural readings leave the device                                                   |                                          |                                                     |
| R12 | Affordable                  | BridgePulse-specific parts \$250 or less, with the complete cost including the FieldNode core stated beside it         | Priced BOM ( bom/ bom.csv )              | At risk: \$248.00 (0.8 % margin); complete \$374.00 |
| R13 | Present results responsibly | Every output labeled as monitoring data that supports inspection; no pass or fail safety rating anywhere in the system | Review of dashboard and report templates | Met by design intent; not yet implemented           |

## ■ Requirements not met or at risk

No requirement is now shown as not met.

- **R2 (repeatability) at risk:** on a light footbridge the mass of each walker lowers the first mode by up to 6.3 %. Gating out the time when someone is on the span meets 0.2 % in simulation, but the footfall model is idealized and the method must be checked on recorded data (TRL 4, on hold).
- **R3 (change detection) not verifiable at TRL 3:** it depends on modeling temperature and any remaining load effect to a daily residual of 0.75 % or less.
- **R10 and R12 at risk:** FieldNode heat and coating life, and a \$2.00 cost margin on indicative prices.
- **R4 and R8 not verifiable at TRL 3:** gauge drift and fitting time need tests or a trial fit.

## ■ Assumptions

- The accelerometer follows the ADXL355 datasheet summary ([Analog Devices](#)); a different part must meet R1.
- The FieldNode core offers 100 mW average for sensors as its design value (115 mW ceiling, FND-CAL-001) and two sealed M12 5-pin ports with a switched rail. BridgePulse assumes RS-485 and a 5 V rail on FieldNode's candidate pinout, which is still open (BRP-DDR-001, O1).
- A temperature-compensated frequency baseline needs at least four weeks of data before any flag, following the approach of [Peeters and De Roeck \(2001\)](#).
- Steel modulus 210 GPa for converting strain to stress. The other calculation assumptions are in BRP-CAL-001, Table 1.